

CHEM 109A

Organic Chemistry

**Wait Lists and Add code requests:
Got you emails, working on it!**

**Questions? (non chemistry)
Chika Anyiwo, undergraduate advisor
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CHEM 109A

Organic Chemistry

**Questions? (chemistry!)
CLAS tutor for this 109A course
Timothy Ngo: txngo@umail.ucsb.edu
Sign up!**

Ch. 1: Important topics/concepts

Hybrid orbitals: s and p orbitals mix to form new hybrid orbitals
(comes from valence-shell electron-pair repulsion (VSEPR) theory
explains geometry of CH₄ and many other molecules

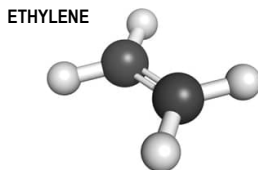
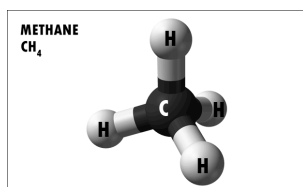
Geometry (shape of molecules):

CH₄: tetrahedral, four sp^3 orbitals, bond angle 109°,

H₂C=CH₂, ethylene, three sp^2 orbitals, one p orbital, bond angle ~120°

HC≡CH, acetylene, two sp bonds, 2 p orbitals, bond angle 180°

Bond length and strength: shorter bonds are stronger.



The Distribution of Electrons in an Atom

quantum
numbers

Table 1.1 Distribution of Electrons in the First Four Shells

		First shell	Second shell	Third shell	Fourth shell
n	Atomic orbitals	s	s, p	s, p, d	s, p, d, f
l	Number of atomic orbitals	1	1, 3	1, 3, 5	1, 3, 5, 7
m_l	Maximum number of electrons	2	8	18	32

2 e per
orbital

- The first shell is closest to the nucleus.
- The closer the atomic orbital is to the nucleus, the lower its energy.
- Within a shell, s lower in energy than p.

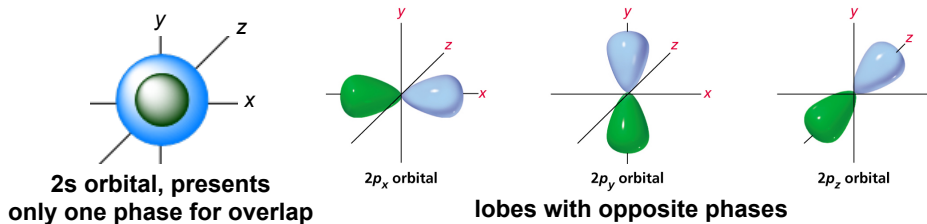
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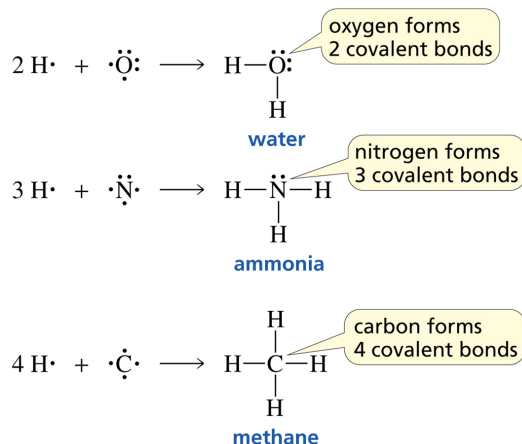


Achieving a Filled Outer Shell by Sharing Electrons



H is surrounded by 2 electrons
Cl is surrounded by 8 electrons

How Many Bonds Does an Atom Form?



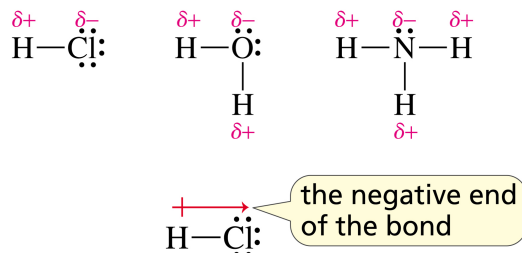
notice that each O, N, C is surrounded by 8 electrons, and each H is surrounded by 2 electrons

Nonpolar and Polar Covalent Bonds

Nonpolar covalent bond = bonded atoms are the same
or have similar electronegativities.



Polar covalent bond = bonded atoms have
different electronegativities.

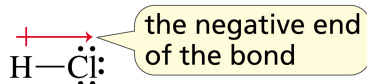
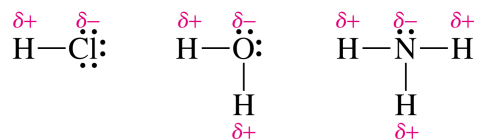


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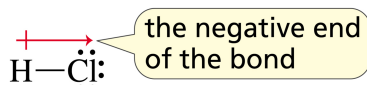
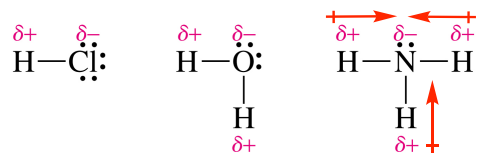


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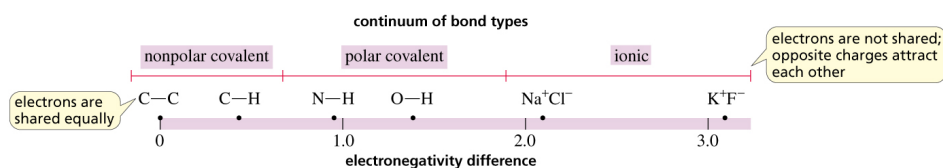
ELECTRONEGATIVITY OF ELEMENTS

Electronegativity increases

Period																																					
1	H 2.20																																				He
2	Li 0.98		Be 1.57																B 2.04		C 2.55		N 3.04		O 3.44		F 3.98		Ne								
3	Na 0.93		Mg 1.31																Al 1.61		Si 1.90		P 2.19		S 2.58		Cl 3.16		Ar								
4	K 0.82	Ca 1.00	Sc 1.36	Ti 1.54	V 1.63	Cr 1.66	Mn 1.55	Fe 1.83	Co 1.88	Ni 1.91	Cu 1.90	Zn 1.65	Ga 1.81	Ge 2.01	As 2.18	Se 2.55	Br 2.96	Kr 3.00																			
5	Rb 0.82	Sr 0.95	Y 1.22	Zr 1.33	Nb 1.6	Mo 2.16	Tc 1.9	Ru 2.2	Rh 2.28	Pd 2.20	Ag 1.93	Cd 1.69	In 1.78	Sn 1.96	Sb 2.05	Te 2.1	I 2.66	Xe 2.6																			
6	Cs 0.79	Ba 0.89	*	Hf 1.3	Ta 1.5	W 2.36	Re 1.9	Os 2.2	Ir 2.20	Pt 2.28	Au 2.54	Hg 2.00	Tl 1.62	Pb 2.33	Bi 2.02	Po 2.0	At 2.2	Rn																			
7	Fr 0.7	Ra 0.9	**	Rf	Db	Sg	Bh	Hs	Mt	Ds	Rg	Uub	Uut	Uuq	Uup	Uuh	Uus	Uuo																			
Lanthanides	*		La 1.1	Ce 1.12	Pr 1.13	Nd 1.14	Pm 1.13	Sm 1.17	Eu 1.2	Gd 1.2	Tb 1.1	Dy 1.22	Ho 1.23	Er 1.24	Tm 1.25	Yb 1.1	Lu 1.27																				
Actinides	**		Ac 1.1	Th 1.3	Pa 1.5	U 1.38	Np 1.36	Pu 1.28	Am 1.13	Cm 1.28	Bk 1.3	Cf 1.3	Es 1.3	Fm 1.3	Md 1.3	No 1.3	Lr 1.3																				

Differences in electronegativity between atoms lead to polar bonds
The bigger the difference, the more polar the bond

The Greater the Difference in Electronegativity, the More Polar the Bond



Nonpolar covalent bond: electronegativity difference < 0.5

Polar covalent bond: electronegativity difference $0.5 - 1.9$

Electronegativity difference > 1.9 : electrons are not shared;
 atoms are held together by
 the attraction of opposite charges

Dipole Moment

Dipole moment = size of the charge $\times$ the distance between the charges

Table 1.4 The Dipole Moments of Some Common Bonds

Bond	Dipole moment (D)	Bond	Dipole moment (D)
H—C	0.4	C—C	0
H—N	1.3	C—N	0.2
H—O	1.5	C—O	0.7
H—F	1.7	C—F	1.6
H—Cl	1.1	C—Cl	1.5
H—Br	0.8	C—Br	1.4
H—I	0.4	C—I	1.2

The greater the difference in electronegativity,
the greater the dipole moment and the more polar the bond.

Dipole Moment

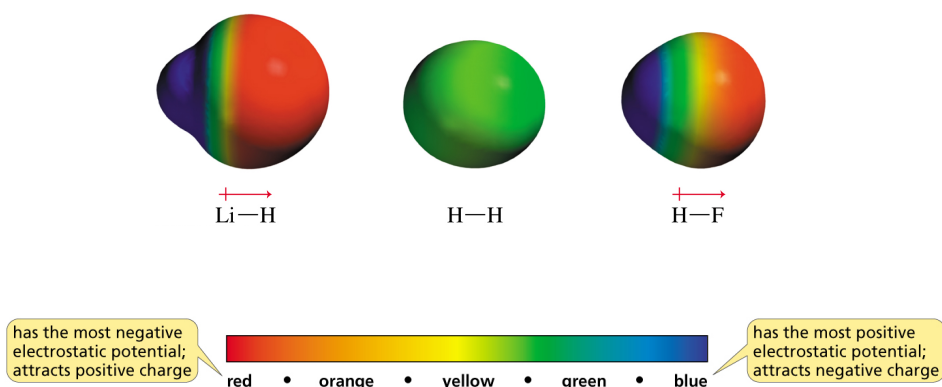
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H—I	0.4	C—I	1.2

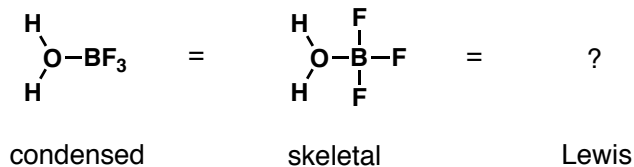
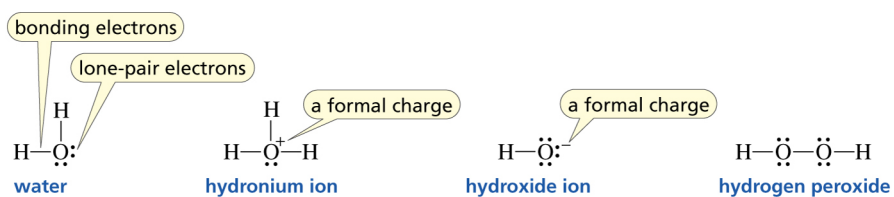
The greater the difference in electronegativity,
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Electrostatic Potential Maps

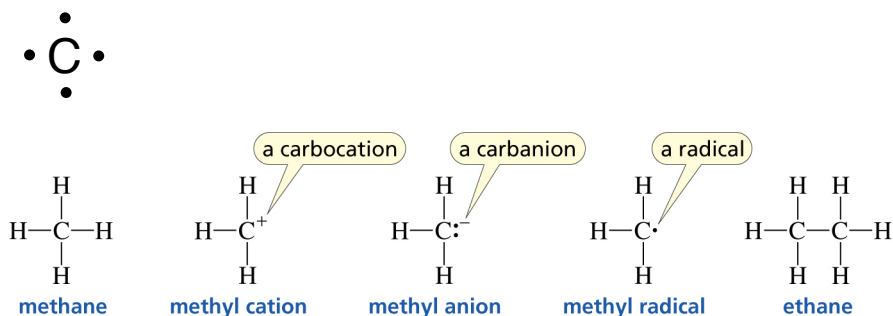


Lewis Structures

Important items: lone pairs, formal charges

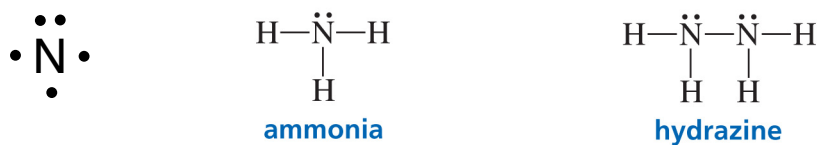


Carbon Forms Four Bonds



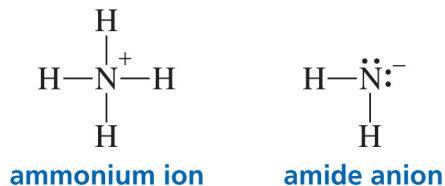
If carbon does **not** form **four** bonds,
it has a **charge** (or it is a **radical**).

Nitrogen Forms Three Bonds

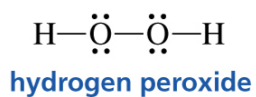
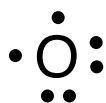


Nitrogen has **one** lone pair.

If nitrogen does **not** form **three** bonds, it is **charged**.

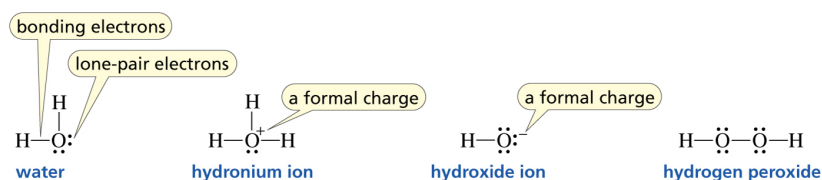


Oxygen Forms Two Bonds

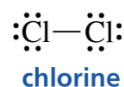
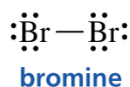


Oxygen has **two** lone pairs.

If oxygen does **not** form **two** bonds, it is **charged**.

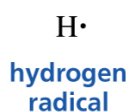
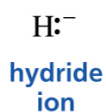


Hydrogen and the Halogens Form One Bond

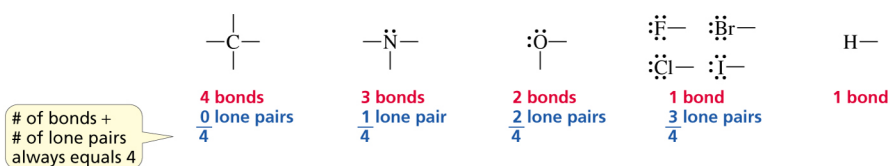


A **halogen** has **three** lone pairs.

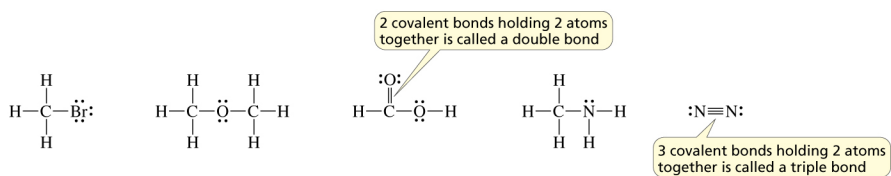
If hydrogen or halogen does **not** form **one** bond, it has a **charge** (or it is a **radical**).



The Number of Bonds Plus the Number of Lone Pairs Equals Four



Lewis Structures



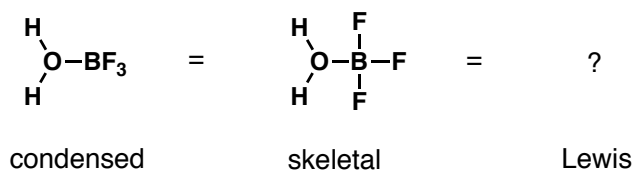
Formal Charge

Important items: lone pairs, formal charges

Formal Charge =

the # of valence electrons –

(the # of lone-pair electrons + the # of bonds)



How to Draw a Lewis Structure

review Gen. Chem. and practice!

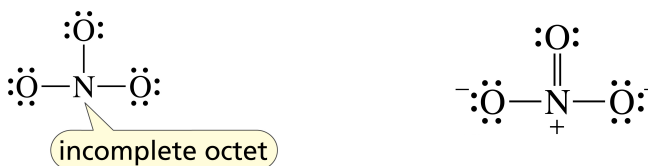
Homework: problems 49, 54, 55

NO_3^- (anion of nitric acid HNO_3)

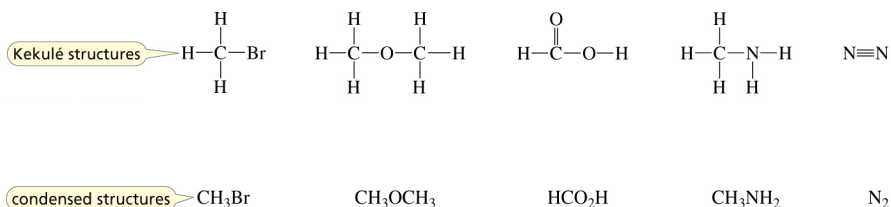
Determine the total number of valence electrons ($5 + 6 + 6 + 6 = 23$). Because they are negatively charged, add another electron = 24.

Avoid O—O bonds.

Check for formal charges.



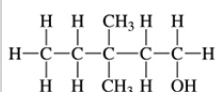
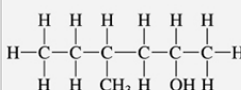
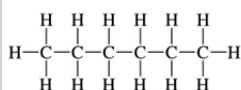
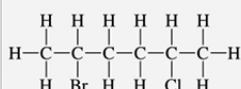
Kekulé Structures and Condensed Structures



Kekulé Structures and Condensed Structures

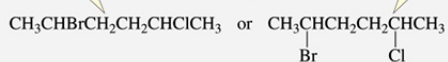
Table 1.5 Kekulé Structures and Condensed Structures

Kekulé Structures



Condensed Structures

atoms bonded to a carbon are shown to the right of the carbon...

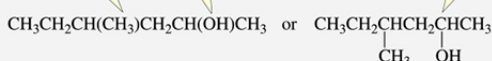


...or connected to a carbon by a line bond above or below the carbon

repeating CH_2 groups can be shown in parentheses

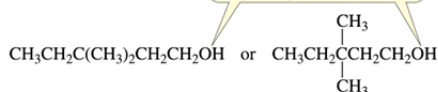


groups bonded to a carbon can be shown in parentheses to the right of the carbon...

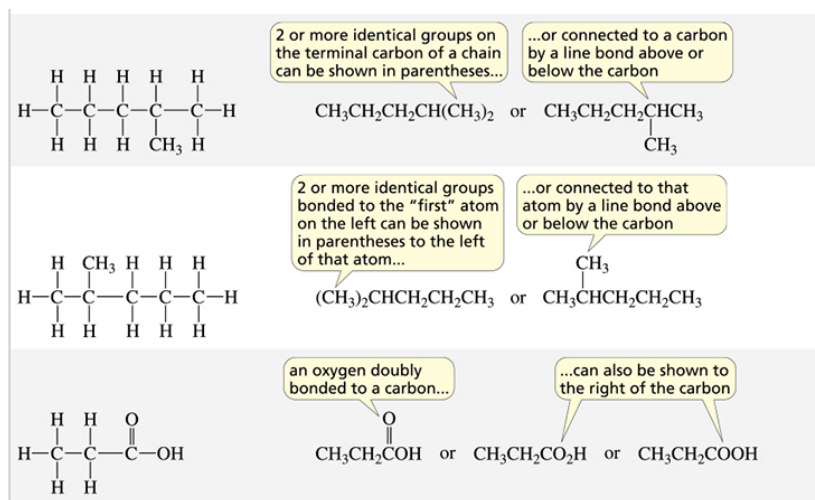


...or connected to the carbon by a line bond above or below the carbon

a single group bonded to the terminal carbon of a chain is not put in parentheses

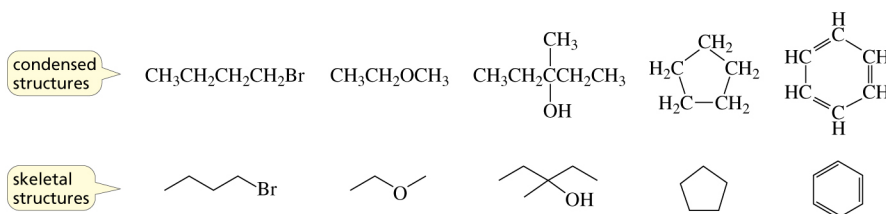


Kekulé Structures and Condensed Structures

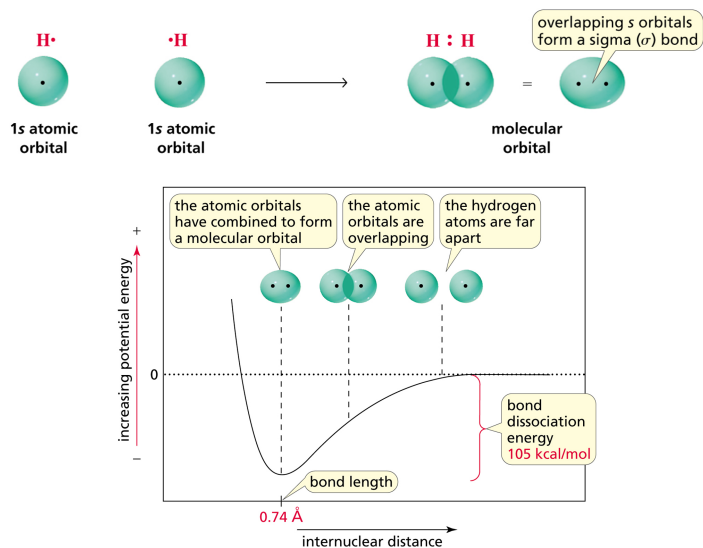


Skeletal Structures

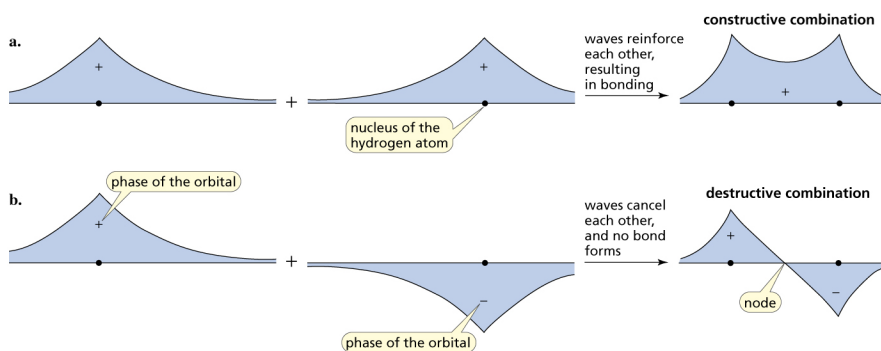
Skeletal structures show the carbon-carbon bonds as lines, but do not show the carbons or the hydrogens that are bonded to the carbons.



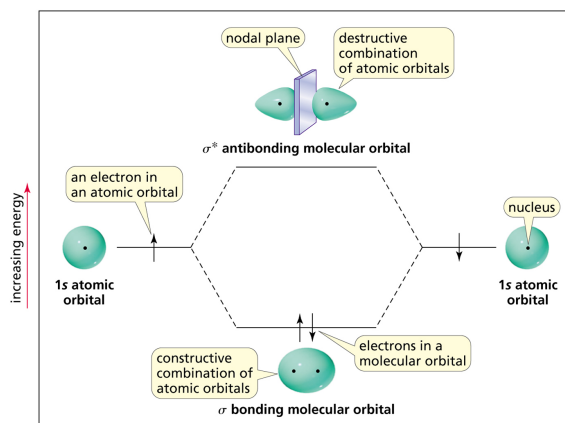
Back to Bonding: Forming a Sigma Bond



Waves Can Reinforce Each Other Waves Can Cancel Each Other



Atomic Orbitals Combine to Form Molecular Orbitals



Orbitals are Conserved

of Molecular Orbitals = # of Atomic Orbitals Combined

Side-to-Side Overlap of In-Phase p Orbitals Forms a π Bond

